Building the Internet of Things

From Global Concepts to Classroom Creation

A visual guide to coding your first smart device

Connecting the Physical and Digital Worlds

The Internet of Things (IoT) is a network of physical objects embedded with sensors and software. These things collect and exchange data over a network without needing human interference.

By 2025, the number of **internet-connected IoT devices** worldwide is predicted to exceed 75 Billion.

Everyday Objects Become Smart

How a simple refrigerator connects to a computer network to improve efficiency via Wi-Fi, Bluetooth, or NFC.

Scans groceries to track inventory.

Displays product expiration dates.

Automatically sends emails and smartphone notifications before food spoils.

The Four Pillars of an IoT System

1. Devices & Sensors

Collect data from the surrounding environment (e.g., taking a temperature reading).

2. Connectivity

Bridges the sensor to the internet or computer using cellular, satellite, Wi-Fi, or Bluetooth.

3. Data Processing

The computer uses software and predefined rules to process the incoming sensor data.

4. User Interface (UI)

Displays results and allows users to remotely control the application via an app.

Project Mission: Become a Creator

Build a custom IoT system to control an LED light remotely using an Android smartphone app.

Phase 1
Wire the Electronic Hardware

Phase 2
Code the Microprocessor Brain

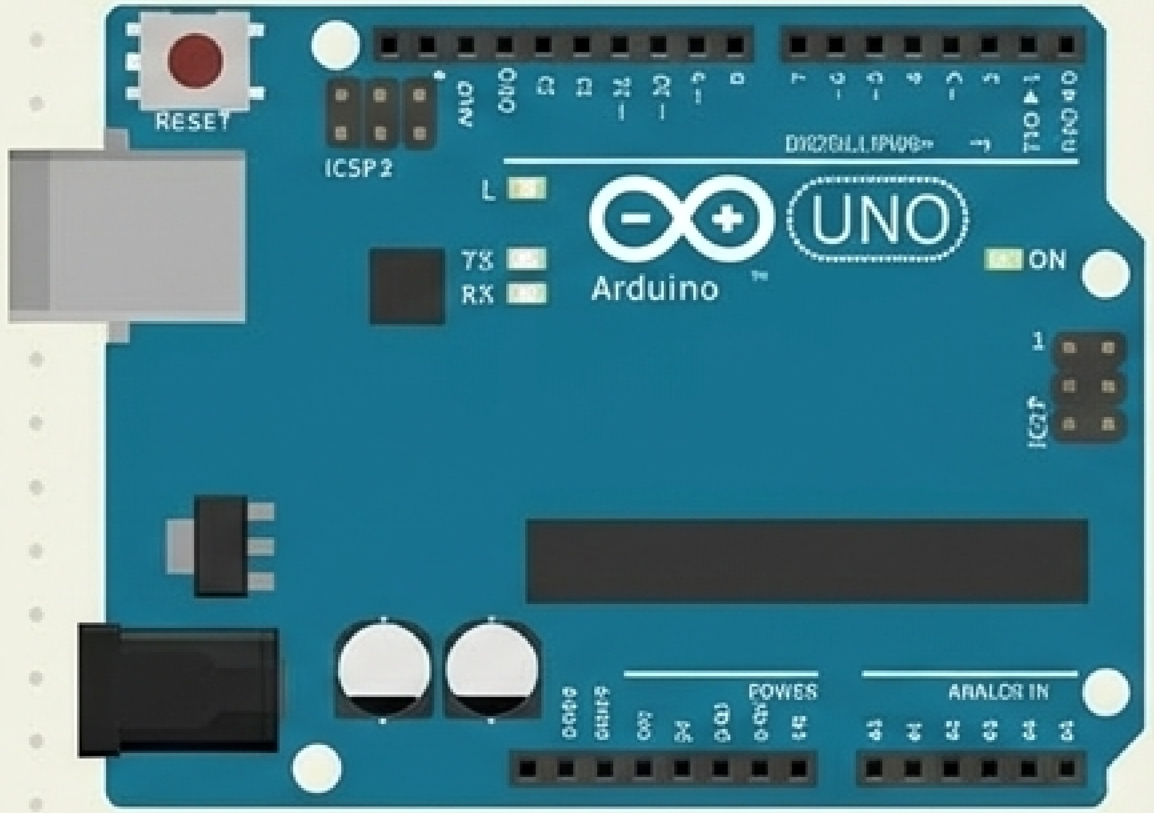
{...;}

Phase 3
Design the App Interface

Phase 4
Connect via Bluetooth

Anatomy of Your Hardware

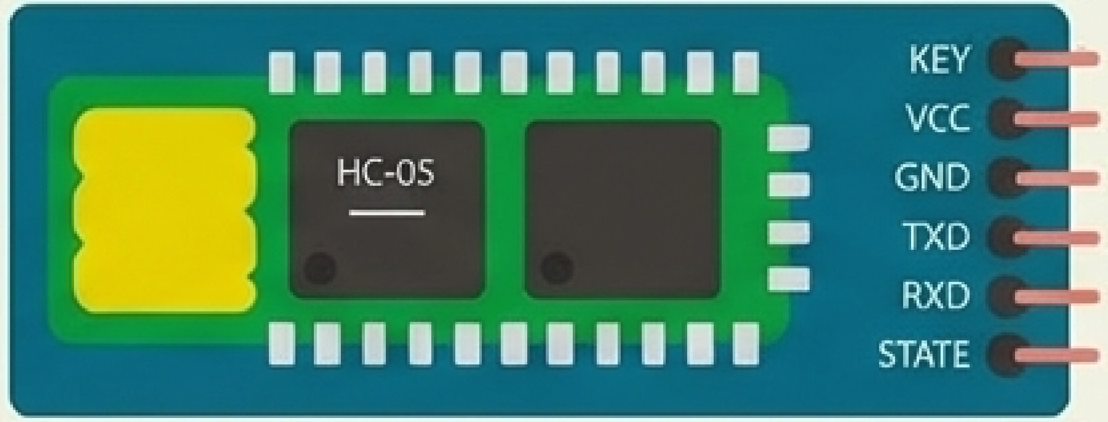
Component 1: Arduino UNO



(The Brain)

An open-source hardware board. It takes inputs from sensors, processes data, and sends outputs to devices.

Component 2: HC-05 Bluetooth Module



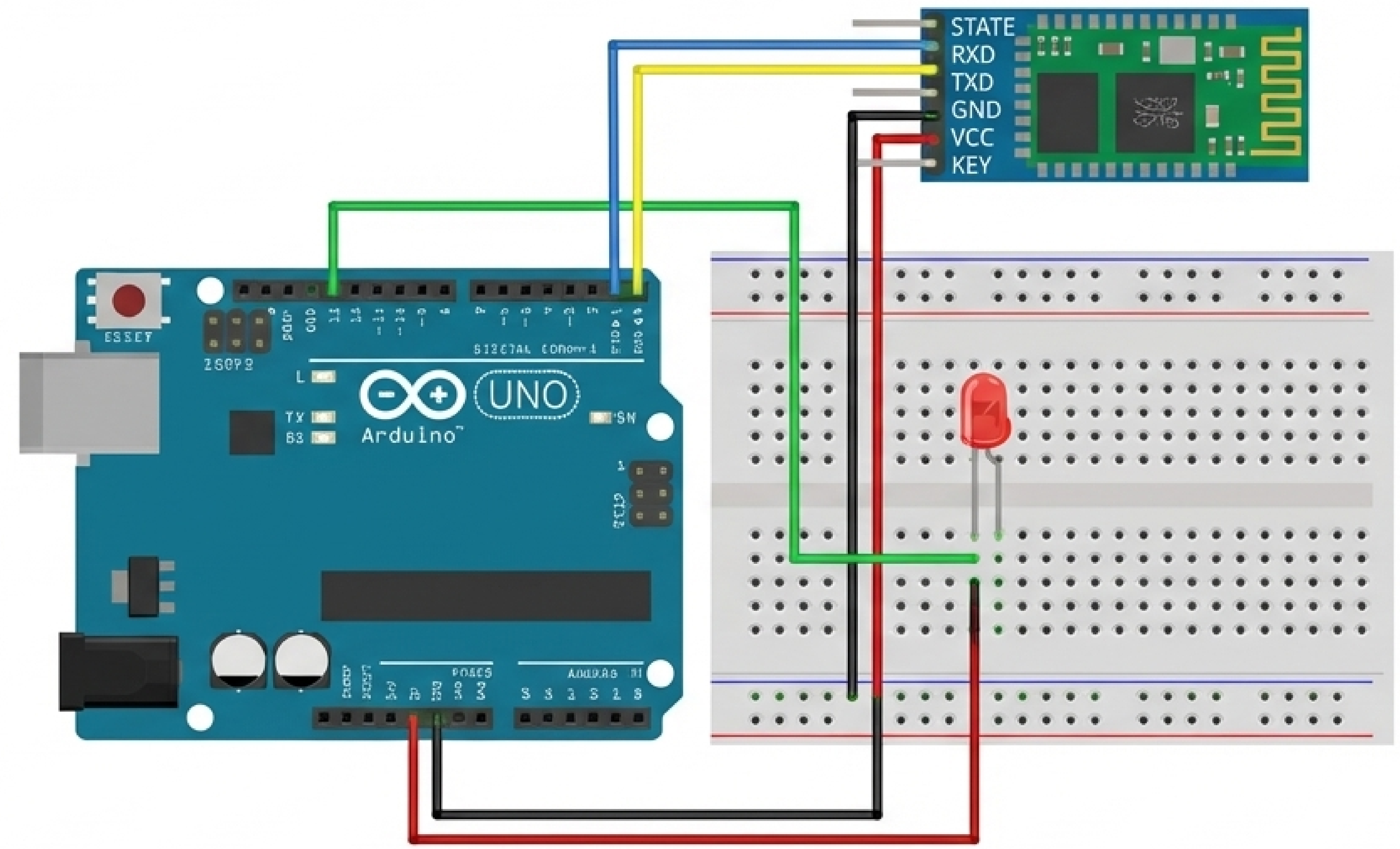
(The Communicator)

A wireless communication device. It allows two processors (like your Arduino and your smartphone) to talk to each other over short distances.

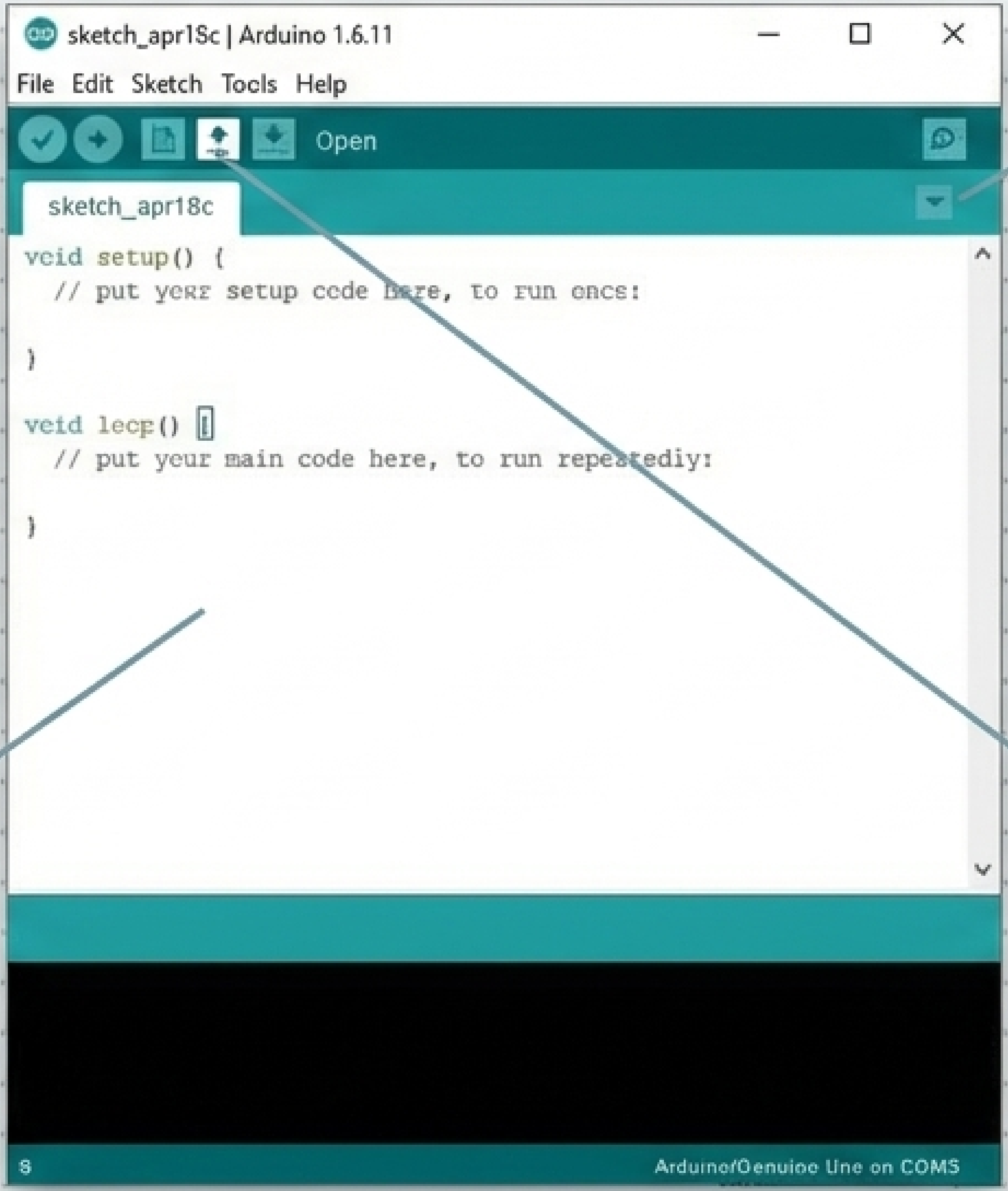
Step 1: Wiring the Circuit Blueprint

Connection Legend

- Arduino pin 0 -> HC05 pin TXD
- Arduino pin 1 -> HC05 pin RXD
- Arduino pin GND -> HC05 pin GND
- Arduino pin 5V -> HC05 pin VCC
- Arduino pin 13 -> LED Anode
- Arduino pin GND -> LED Cathode



The Software Toolkit



The Environment

Arduino IDE (Integrated Development Environment) is the software used to write and upload instructions to the board.

The Language

Programs are written in C++ (with added methods and functions).

The Canvas

A program written in this language is called a 'sketch' and is saved with an .ino file extension.

NotebookLM

Step 2: Coding the Brain (Setup)

```
void setup()
{
  Serial.begin(38400);
  pinMode(13, OUTPUT);
}
```

Instructions that run only once when the device powers on.

Sets the communication speed (baud rate) for the Bluetooth module.

Tells the Arduino that Pin 13 will be used to send power OUT to the LED.

Step 2: Coding the Logic (The Loop)

```
if(inputString == "1") {
  digitalWrite(13, HIGH);
} else if(inputString == "0") {
  digitalWrite(13, LOW);
}
```

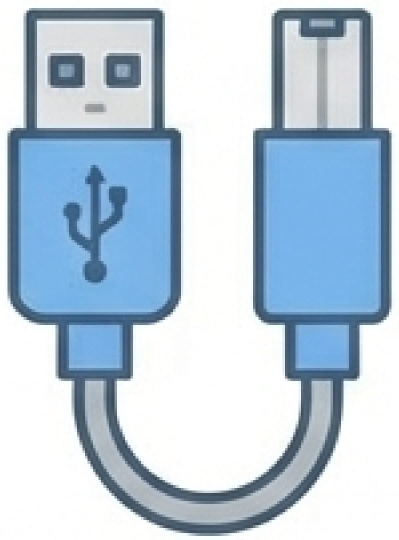
Instructions that run continuously, waiting for input.

If the Bluetooth module hears a '1' from the phone...

...turn the LED **ON**.

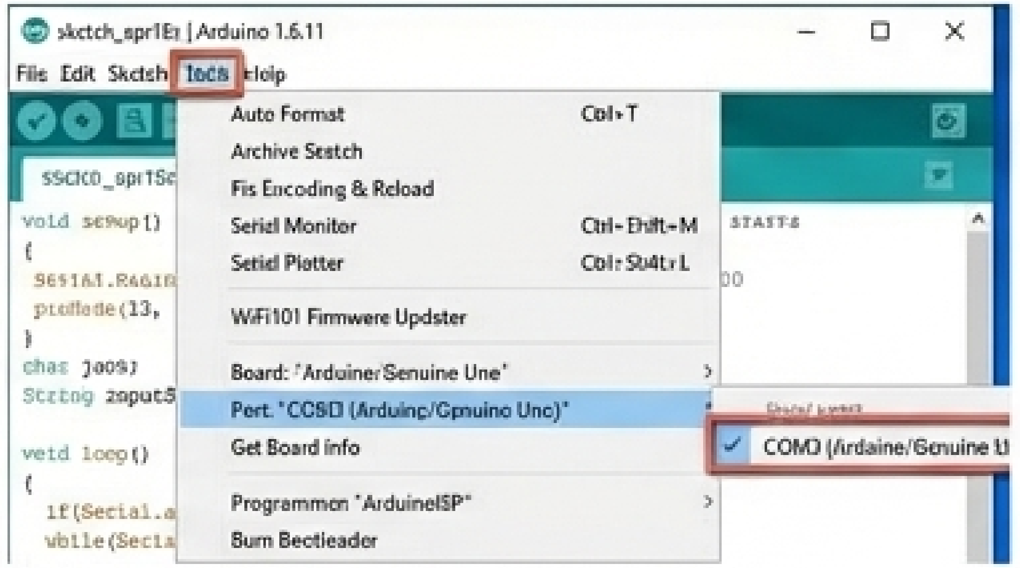
...turn the LED **OFF**.

Uploading the Instructions



Step 1: Connect

Plug the Arduino Uno board into the computer using a USB cable.



Step 2: Select Board

From the Tools menu, select 'Arduino/Genuino Uno'. Then, select the specific COM port.

Step 3: Upload

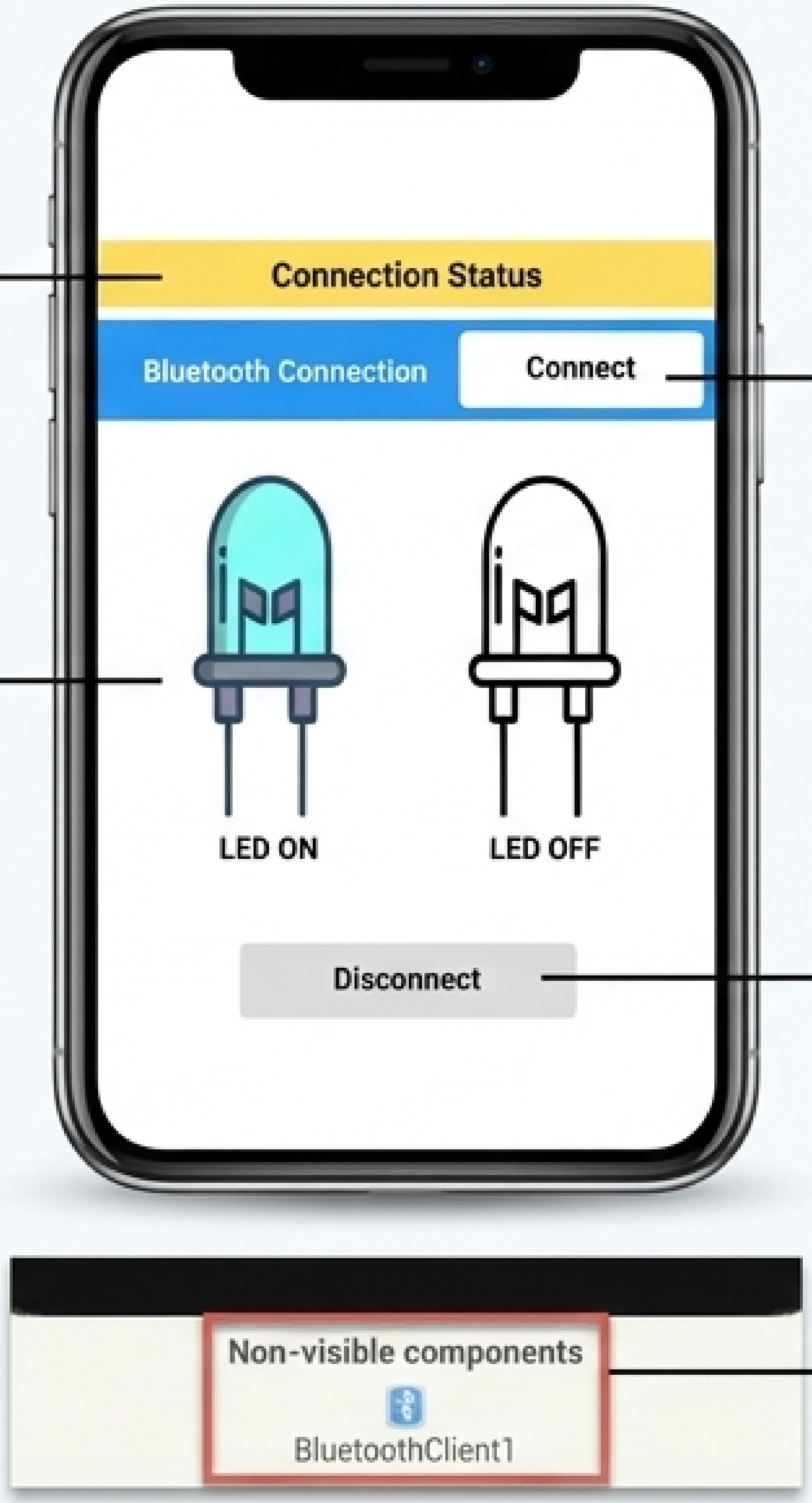
Click the right-facing arrow Upload button to send the sketch into the Arduino's memory.

Step 3: Designing the Remote Control

Platform: MIT App Inventor 2

Status Bar: Displays connection status.

LED Toggle Buttons: Large visual buttons for ON and OFF.



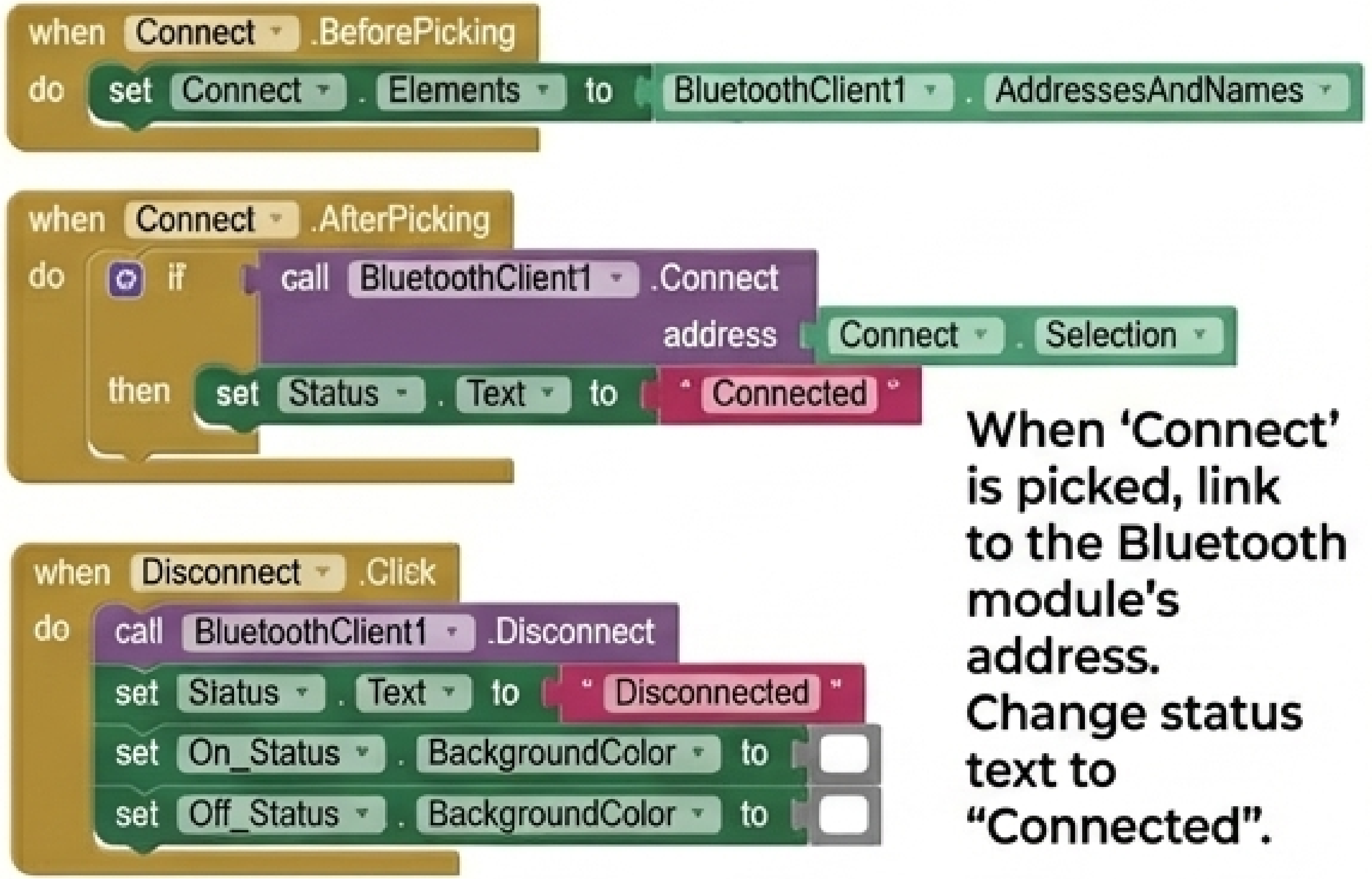
Connect/Disconnect Buttons: To link to the HC-05 module.

Disconnect Buttons.

Non-Visible Component: “BluetoothClient1” running in the background to handle the wireless signal.

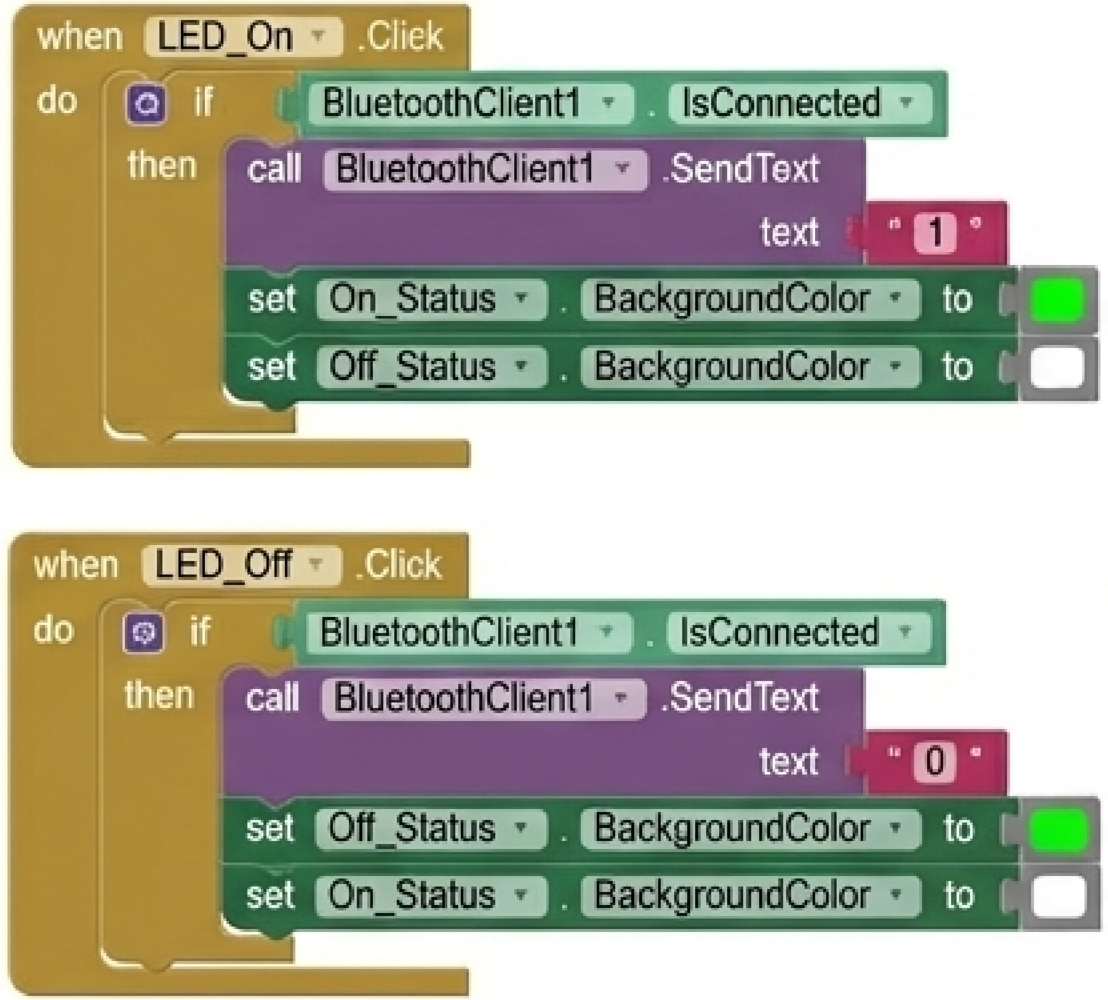
Step 4: Programming the App Logic

Connection Logic



When ‘Connect’ is picked, link to the Bluetooth module's address. Change status text to “Connected”.

Command Logic

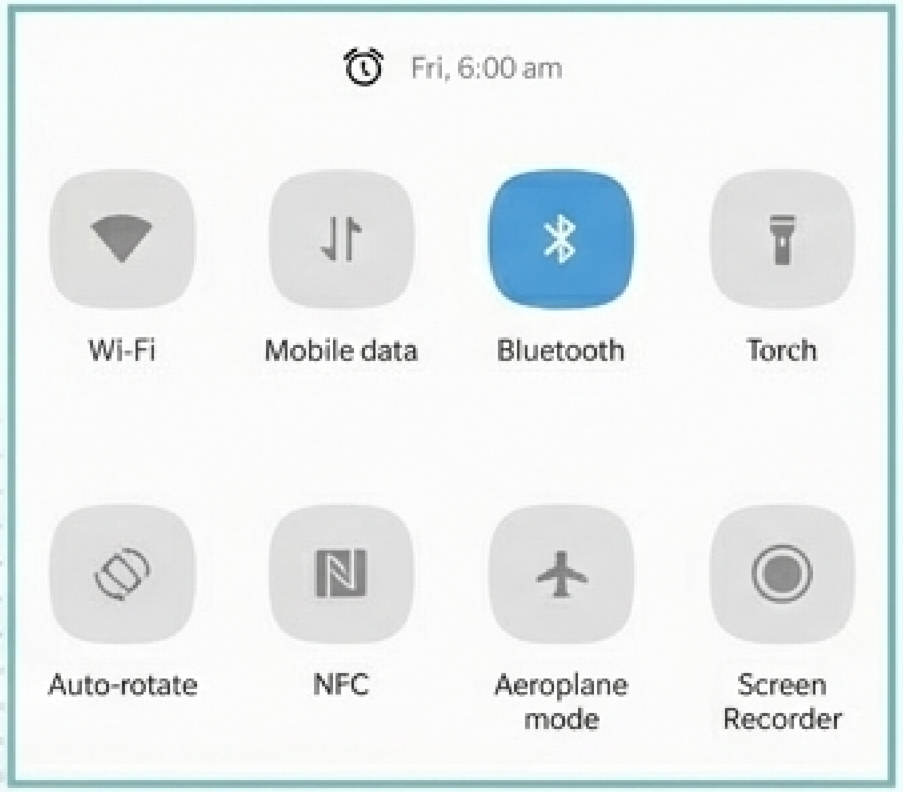


When an LED button is clicked, check if Bluetooth is connected.

Send the text ‘1’ or ‘0’ over the air. Change the button background color to show it is active.

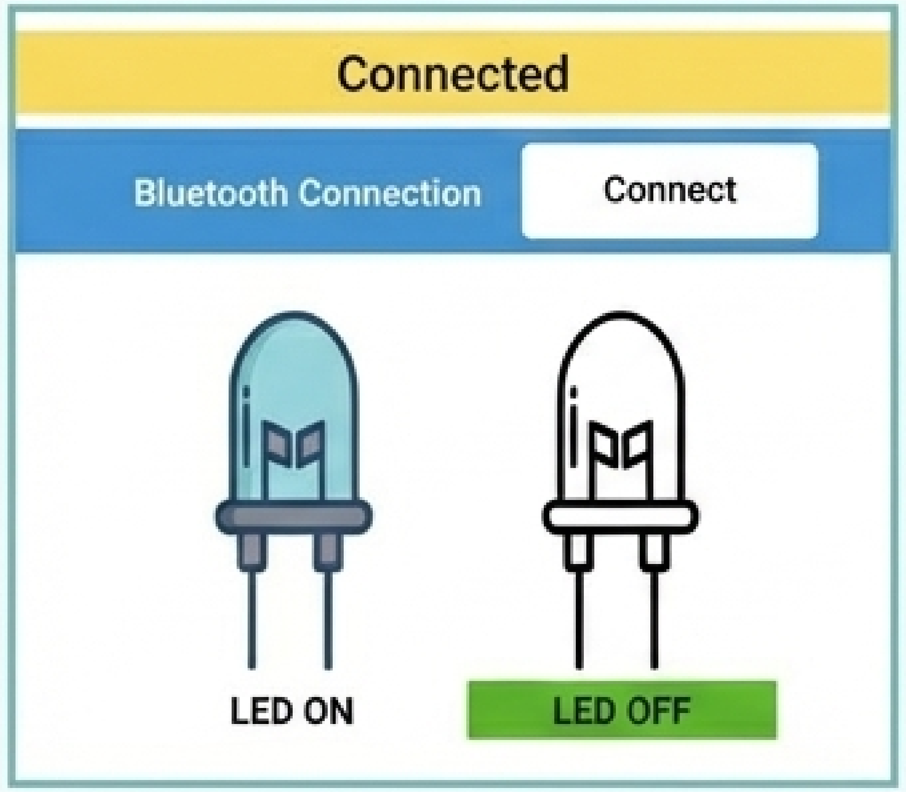
Step 5: Connecting and Testing

Pairing



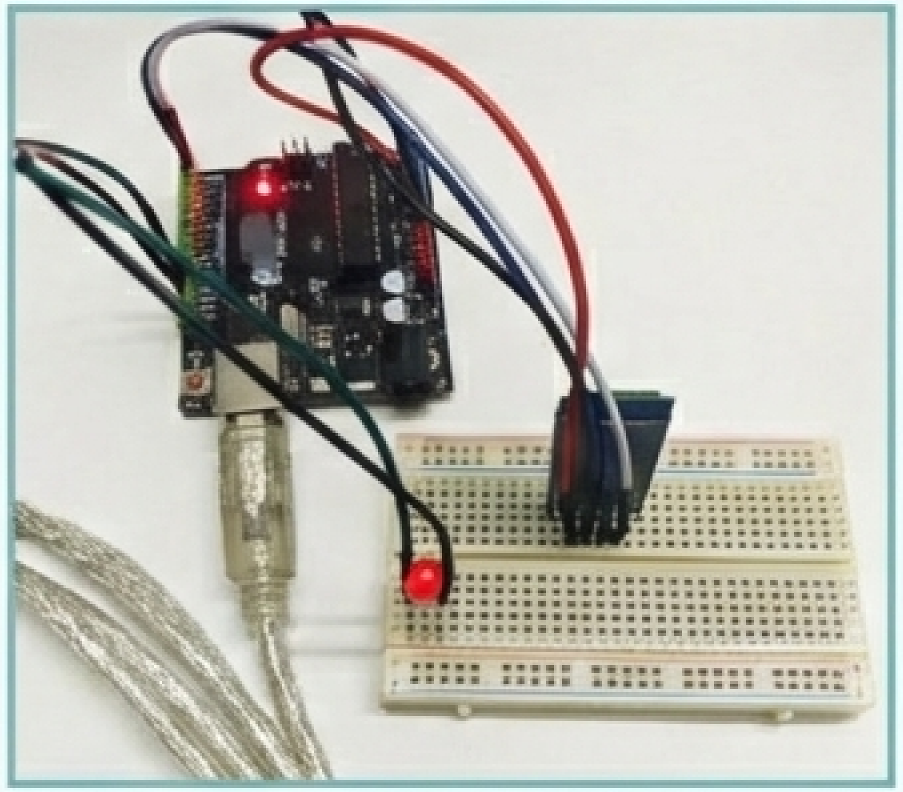
Open phone settings, enable Bluetooth, and pair with the HC-05 module network.

Activation



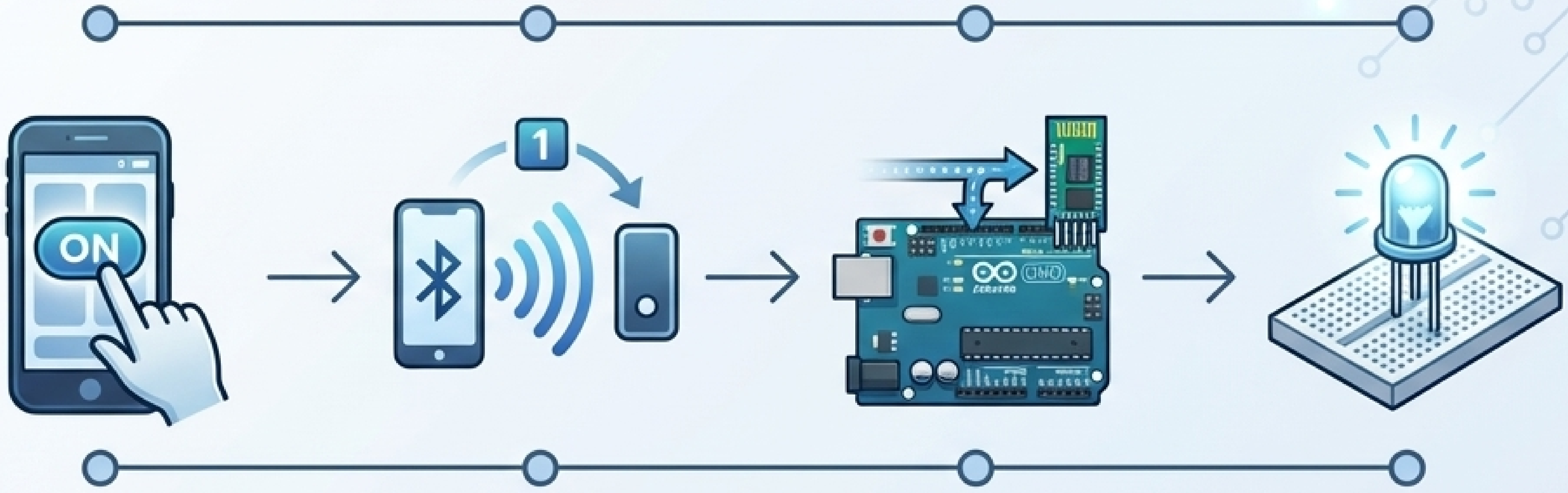
Open the app, press 'Connect', and tap the LED ON or OFF buttons.

The Result



Watch the physical hardware respond instantly to your wireless commands.

Synthesis: How the Data Flows



User Interface	Connectivity	Processing	Device
Finger taps ‘ON’ on the App.	App sends a text ‘1’ via Bluetooth waves.	HC-05 receives ‘1’, Arduino reads it and executes digitalWrite(HIGH).	Pin 13 sends 5 Volts, LED illuminates.

Tech Flash

- **IoT:** Network of physical objects exchanging data.
- **Bluetooth:** Open wireless tech for short-distance data transfer.
- **Arduino IDE:** Text-based software to program boards.